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Review Cytoskeleton (Hoboken). 2013 Jul;70(7):360-84. doi: 10.1002/cm.21117.
Epub 2013 Jun 27.

Gelsolin: the tail of a molecular gymnast

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PMID: 23749648 DOI: 10.1002/cm.21117

Abstract

Gelsolin superfamily members are Ca(2+) -dependent, multidomain regulators of the actin cytoskeleton. Calcium binding activates gelsolin by inducing molecular gymnastics (large-scale conformational changes) that expose actin interaction surfaces by releasing a series of latches. A specialized tail latch has distinguished gelsolin within the superfamily. Active gelsolin exhibits actin filament severing and capping, and actin monomer sequestering activities. Here, we analyze a combination of sequence, structural, biophysical and biochemical data to assess whether the molecular plasticity, regulation and actin-related properties of gelsolin are also present in other superfamily members. We conclude that all members of the superfamily will be able to transition between a compact conformation and a more open form, and that most of these open forms will interact with actin. Supervillin, which lacks the severing domain 1 and the F-actin binding-site on domain 2, is the clear exception. Eight calcium-binding sites are absolutely conserved in gelsolin, adseverin, advillin and villin, and compromised to increasing degrees in CapG, villin-like protein, supervillin and flightless I. Advillin, villin and supervillin each contain a potential tail latch, which is absent from CapG, adseverin and flightless I, and ambiguous in villin-like protein. Thus, calcium regulation will vary across the superfamily. Potential novel isoforms of the superfamily suggest complex regulation at the gene, transcript and protein levels. We review animal, clinical and cellular data that illuminate how the regulation of molecular flexibility in gelsolin-like proteins permits cells to exploit the force generated from actin polymerization to drive processes such as cell movement in health and disease.

Keywords: CapG; adseverin; flightless; gelsolin; villin.

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